SHAI SHARATH VIJEY S

AI Software Engineer | Machine Learning, RAG & Data Science | Python, FastAPI, PyTorch

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PROFESSIONAL SUMMARY

Results-driven AI Software Engineer with hands-on experience building production RAG (Retrieval-Augmented Generation) systems, GenAI/LLM (Large Language Model) applications, and Model Context Protocol (MCP) tool integrations, backed by a strong foundation in machine learning, GANs-based synthetic data generation, fraud detection, and exploratory data analysis. Proven track record delivering scalable AI systems using Python, FastAPI, and vector databases (Qdrant, ChromaDB). Seeking an AI/ML Engineering role to build impactful, production-grade AI products.

TECHNICAL SKILLS

Programming & Analysis: Python, SQL, Pandas, NumPy, Scikit-learn, PyTorch

AI / ML: Supervised & Unsupervised Learning, GANs (CTGAN, RealTabFormer), Predictive Modeling, RAG (Retrieval-Augmented Generation), Generative AI (GenAI), Large Language Models (LLM), Model Context Protocol (MCP)

Vector Databases & Search: Qdrant, ChromaDB, Semantic Search, Hybrid Retrieval (BM25 + Dense Embeddings), Cross-Encoder Reranking

Data Handling: Data Cleaning, Feature Engineering, Statistical Analysis

Visualization: Power BI, Matplotlib, Seaborn

Backend & Deployment: Flask, FastAPI, Gunicorn, Render, REST APIs, Async Programming (asyncio, httpx)

Automation & Tooling: Selenium Web Scraping, PyInstaller (EXE Packaging), FastMCP

Other: Agile, Git, SDLC, EDA, Problem Solving, MS Excel (VLOOKUP, Pivot Tables)

Soft Skills: Written & Oral Communication, Analytical Thinking, Decision Making, Time Management, Attention to Detail, Team Collaboration

WORK EXPERIENCE

Junior AI Software Engineer | Adela Software and Services Pvt. Ltd., Trichy, India *Sept 2025 - Present*

- Engineered core components of a production RAG (Retrieval-Augmented Generation) platform using FastAPI, ChromaDB, Qdrant, and LangChain, implementing hybrid retrieval (BM25 + dense embeddings) and cross-encoder reranking for natural-language tax-form queries.
- Delivered key modules of a GPU-orchestrated model pool for scalable multi-worker LLM inference, an LLM-driven query analysis system, and a multi-session AI chat service.
- Built critical components of a static-analysis engine and Qdrant-based vector search system for a legacy, undocumented PHP/ASP codebase, exposed via a Flask analytics dashboard with async job processing and drift detection.

Artificial Intelligence Intern | CloudplusAI Technology, Trichy, India *Apr 2025 - May 2025*

- Assisted in AI research and development aligned with company projects.
- Contributed to team discussions and applied theoretical knowledge effectively.

Machine Learning Intern | Testaing Solutions, Bangalore, India *Nov 2024 - Feb 2025*

- Developed and tested GANs (CTGAN, RealTabFormer) for synthetic data generation.
- Conducted data preprocessing, EDA, and statistical validation across finance and healthcare domains.
- Improved test data management by integrating model-based synthetic data pipelines; built dashboards comparing real vs. synthetic datasets.

Machine Learning Intern | Verzeo, Bangalore, India *Nov 2022 - Dec 2022*

- Worked on ML model development and data preprocessing for real-time use cases.
- Gained hands-on experience with machine learning and collaborative project execution.

KEY PROJECTS

AI-Powered Tax Form Processing & RAG Platform | Adela Software and Services

Tech: FastAPI, LangChain, ChromaDB, Qdrant, Hybrid Search, GPU Model Orchestration

- Engineered core retrieval and reranking components of a production RAG system combining hybrid search, cross-encoder reranking, and GPU-orchestrated multi-worker inference to answer natural-language queries mapped to reference tax forms and publications.

Legacy PHP Codebase Intelligence & Vector Search Dashboard | Adela Software and Services

Tech: Python, Flask, Qdrant, Regex-based Static Analysis, JavaScript

- Built key components of a system that reverse-engineered an undocumented legacy PHP/ASP application, including a vector-indexed code-chunk search system and a Flask dashboard with async build pipelines and drift detection.

MCP Tool Hub - Multi-Tool AI Assistant Platform | Personal Project

Tech: Python, FastMCP (Model Context Protocol), Flask, Selenium, httpx, PyInstaller

- Designed and built a modular AI tool server on the Model Context Protocol (MCP) with 10+ callable tools: real-time weather/rain detection, currency conversion, a safe math-expression calculator, multi-provider text translation, QR code generation, image format conversion, English dictionary lookup, timezone conversion, and Google Maps/News web scraping via Selenium.

- Built a dynamic tool-loading system that auto-discovers and registers tool modules at runtime, plus a Flask web UI that parses natural-language commands and routes them to the correct MCP tool.
- Developed an automated EXE packaging pipeline using PyInstaller that auto-detects project dependencies and hidden imports to build a standalone Windows executable from any Flask/MCP project.

Synthetic Data Generation using GAN | Academic Project

- Built a GAN-based system (CTGAN + custom GAN) to generate, evaluate, and download synthetic tabular data, achieving 90% statistical similarity to real datasets. Tools: Flask, PyTorch, Pandas, Matplotlib, Seaborn.

EDUCATION

B.Tech, Artificial Intelligence and Data Science - K. Ramakrishnan College of Technology, Tamil Nadu | CGPA: 7.3 | 2021 - 2025

HSC - Sowdambikaa Matric Boys Higher Secondary School | 87.02% | 2020 - 2021

SSLC - Sowdambikaa Matric Boys Higher Secondary School | 84.6% | 2018 - 2019

CERTIFICATIONS

GenAI Powered Data Analytics Job Simulation - Tata & Forage

Machine Learning, Deep Learning - UpGrad

Database Management System, Python, Foundation of Data Science - UpGrad

Foundations of Data Science - Scaler

Data Science Program - LMES Academy & Uptor

ACHIEVEMENTS

Won Regional Hackathon at PSG College of Technology (Low-Budget Air Purifier Model)

Won inter- and intra-college Talent Show (Singing)

LANGUAGES & INTERESTS

Languages: English (Fluent), Tamil (Fluent)

Interests: AI & Machine Learning, Data Science, Open Source Projects, Blogging